

What to Measure

Optimal Monitoring under Adverse Selection and Moral Hazard

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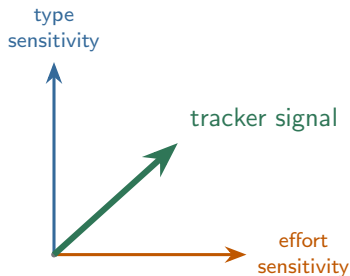
The question

An insurer offers coverage to a driver:

- the driver's **ability** θ is privately known;
- the driver chooses **care** a , which the insurer cannot observe.

Accident records reflect both. A **driving tracker** can provide additional information.

With limited monitoring capacity, should the tracker measure **driving ability**, **driving behavior**, or both?



The insurer chooses the tracker's **sensitivity to effort and type**.

Monitoring often serves two purposes

In many applications, the same monitoring system provides information about both **type** and **effort**.



Insurance

driver risk



driving behavior



Workplace

worker ability



worker effort



Lending / platforms

borrower risk



repayment behavior

Across these applications, the design of monitoring is a **general and economically consequential choice** for the principal.

Where this paper sits

Value of information in contracting

Holmström (1979); Shavell (1979); Gjesdal (1982); Jewitt (1988); Kim (1995); Chaigneau–Edmans–Gottlieb (2019).

This paper: Characterize which **additional information** is most valuable, given what existing monitoring already reveals.

Adverse selection with moral hazard

Guesnerie–Laffont (1984); Laffont–Tirole (1986, 1993); Picard (1987); McAfee–McMillan (1987); Gottlieb–Moreira (2022); Castro-Pires et al. (2024).

This paper: Price local screening and effort incentives jointly to characterize the **demand for monitoring** they generate.

Monitoring design and strategic signals

Georgiadis–Szentes (2020); Li–Yang (2020); Frankel–Kartik (2019); Ball (2025); Hörner–Lambert (2021).

This paper: Derive the optimal measurement direction under scarce capacity, including when the signal responds to **type** and **effort** in **opposite directions**.

Model: primitives

- **Type** The agent privately observes $\theta \in [\underline{\theta}, \bar{\theta}]$, $\theta \sim F$.
- **Effort** The agent chooses unobservable effort $a \in [\underline{a}, \bar{a}]$ at cost $c(\theta, a)$.
- **Outcomes** The principal observes outcomes $y \in \mathbb{R}^K$

$$y = B \begin{pmatrix} a \\ \theta \end{pmatrix} + \varepsilon, \quad \varepsilon \sim N(0, \Sigma).$$

The paper allows a general distribution $q(y \mid \theta, a)$.

- **Signal** The principal can design an additional payoff-irrelevant signal $s \in \mathcal{S}$.
- **Preferences** With transfer $t \geq 0$, $u_A = \sqrt{t} - c(\theta, a)$ and $u_P = b(y) - t$.
For simplicity, let principal's gross benefit $b(y) = \gamma^\top y$, where $\gamma \in \mathbb{R}^K$ gives the principal's weights on the observed outcomes.

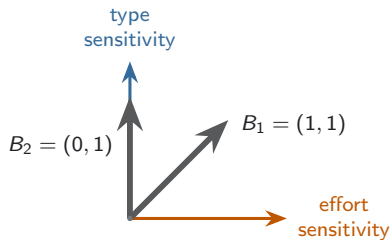
A simple information structure: $K=2$

Collect the two hidden dimensions in x and let the existing performance measures be linear and noisy:

$$x = \begin{pmatrix} a \\ \theta \end{pmatrix}, \quad \boxed{y = Bx + \varepsilon}, \quad \varepsilon \sim N(0, \Sigma).$$

- B : what the existing measures respond to.
- Σ : how noisy they are.
- Example:

$$\text{If } B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \Sigma = I_2, \text{ then}$$
$$y_1 = a + \theta + \varepsilon_1, \quad y_2 = \theta + \varepsilon_2.$$



Each row of B is one measurement direction.
Here B_1 responds to both; B_2 only to type.

What can the principal design?

She adds *one* new signal. Its direction is

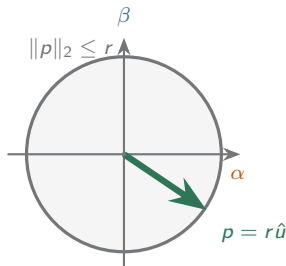
$$p = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \|p\| \leq r$$

and it responds to the state only through the scalar index

$$z = p^T x = \alpha a + \beta \theta,$$

$$\Pr(s = j \mid a, \theta) = g_j(z).$$

α is the signal's sensitivity to **effort**, β its sensitivity to **type**.



The L^2 capacity constraint **does not favor any direction**.

Where should the **first units** of monitoring capacity go? $\lim_{r \downarrow 0} \frac{p^*(r)}{r} = ?$

The direct mechanism

By the generalized revelation principle, we can restrict, without loss, to *direct, obedient* mechanisms

$$\mathcal{M} = \left(\underbrace{a(\hat{\theta})}_{\text{recommended effort}}, \underbrace{t_{\hat{\theta}}(y, s)}_{\text{transfer schedule}} \right).$$

Timing. θ realized $\rightarrow$ report $\hat{\theta} \rightarrow$
recommendation $a(\hat{\theta}) \rightarrow$ *private* effort $\hat{a} \rightarrow$
 $(y, s) \sim P_{\theta, \hat{a}} \rightarrow$ pay $t_{\hat{\theta}}(y, s)$.

$$U(\theta, \hat{\theta}, \hat{a}) = \mathbb{E}_{P_{\theta, \hat{a}}} [u(t_{\hat{\theta}}(y, s))] - c(\theta, \hat{a}).$$

$$(\text{IR}) \quad U(\theta, \theta, a(\theta)) \geq 0,$$

$$(\text{IC}) \quad U(\theta, \theta, a(\theta)) \geq U(\theta, \hat{\theta}, \hat{a}) \quad \forall \hat{\theta}, \hat{a}.$$

One constraint, three deviations

- $\hat{a} \neq a(\theta)$, $\hat{\theta} = \theta$: *obedience* (moral hazard).
- $\hat{\theta} \neq \theta$, $\hat{a} = a(\hat{\theta})$: *truth-telling* (adverse selection).
- both free: **joint deviation**.

The principal's monitoring-design problem

Contracting value at a fixed technology $p = (\alpha, \beta)^T$:

$$V^F(p) = \max_{\mathcal{M}} \mathbb{E}[b(y) - t] \quad \text{s.t.} \quad (\text{IR}), \underbrace{(\text{LL})}_{t \geq 0}, \underbrace{(\text{IC})}_{\text{full joint}}.$$

Monitoring choice:

$$\max_p V^F \quad \text{s.t.} \quad \|p\|_2 \leq r, \quad \text{optimum } p^{F,*}(r) = (\alpha^{F,*}(r), \beta^{F,*}(r))^T.$$

The superscript F denotes objects associated with the full-IC problem.

Which problem is “more important”? — a revealed measure

Not the cost term, or multiplier. It is:

In which direction does the principal spend marginal monitoring capacity?

The direction $(\alpha^*(r), \beta^*(r))$ aggregates all direct, incentive, and joint-deviation effects.

Local incentives: two deviations, one requirement

Let $v = \sqrt{t}$, and let ℓ_a, ℓ_θ be the scores: how observed data respond locally to **effort** and **type**. Let a differentiable $U(\theta)$ denotes the equilibrium utility promised to type θ .

$$\ell(y, s \mid \theta, a; p) = \begin{pmatrix} \ell_a(y, s \mid \theta, a; p) \\ \ell_\theta(y, s \mid \theta, a; p) \end{pmatrix} = \begin{pmatrix} \partial_a \log \phi(y, s \mid \theta, a; p) \\ \partial_\theta \log \phi(y, s \mid \theta, a; p) \end{pmatrix}.$$

Effort deviation

$$\mathbb{E}[v \ell_a] = c_a$$

obedience / moral hazard

Report deviation

$$\mathbb{E}[v \ell_\theta] = U' + c_\theta$$

screening / envelope

$$\ell = \begin{pmatrix} \ell_a \\ \ell_\theta \end{pmatrix}, \quad \Delta(\theta) = \begin{pmatrix} c_a \\ U' + c_\theta \end{pmatrix}, \quad \boxed{\mathbb{E}[v \ell] = \Delta(\theta)}.$$

Adverse selection and moral hazard are economically different, but locally they are two components of one information requirement.

How information prices incentives

Fix an allocation $(a(\cdot), U(\cdot))$ and a type. Suppose the observed data have Fisher information matrix I for (a, θ) . Now $E[v] = U + c$.

Among all transfer schedules satisfying

$$\mathbb{E}[v \ell] = \Delta(\theta),$$

to minimize the expected transfer $E[v^2]$, the cheapest one has cost

$$\boxed{\Delta(\theta)^\top I^{-1} \Delta(\theta)}.$$

The cheapest pay schedule is the projection onto the score space:

$$v^* = \bar{v} + \Delta(\theta)^\top I^{-1} \ell.$$

Incentive requirements are weighted by **inverse information**.

More information in a direction makes incentives in that direction cheaper.

What does the existing system know?

For the existing outcomes

$$y = Bx + \varepsilon, \quad \varepsilon \sim N(0, \Sigma),$$

the information about (a, θ) is

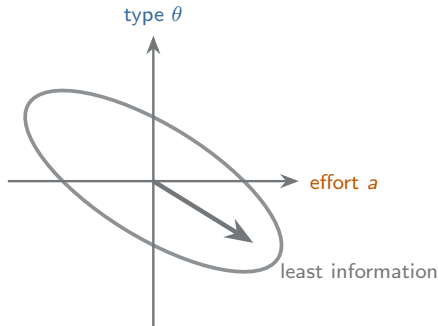
$$I^y = B^\top \Sigma^{-1} B.$$

In the running example,

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Sigma = I_2,$$

so

$$I^y = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad (I^y)^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$



The long axis is the direction the existing data resolve worst.

Information-adjusted incentive needs

Combine what incentives require with what existing monitoring supplies:

$$w(\theta) = (I^y)^{-1} \Delta(\theta).$$

$$\Delta(\theta)$$

incentive needs

$$(I^y)^{-1}$$

inverse existing information

$$w(\theta)$$

what remains valuable to distinguish

In the running example,

$$w = \begin{pmatrix} 2\Delta_a - \Delta_\theta \\ \Delta_\theta - \Delta_a \end{pmatrix}.$$

The optimal signal targets **information-adjusted** needs, not raw incentive needs.

A new signal adds information in one direction

The new signal depends on the state only through

$$z = p^\top x.$$

Its score is always proportional to p , so the information it adds is rank one:

$$I^s(p) = I_g(0) pp^\top + o(r^2).$$

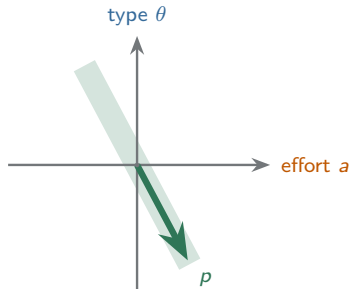
In the Gaussian language, if

$$s = p^\top x + \eta, \quad \eta \sim N(0, \sigma_s^2),$$

then

$$I^s = \frac{pp^\top}{\sigma_s^2}.$$

Signal sensitivity is order r ; the information it adds is order r^2 .



All the added information lies on this line; the design chooses *which* line.

From one type to many types

For one type, the leading gain from direction p is $I_g(0)(p^\top w(\theta))^2 + o(r^2)$, hence

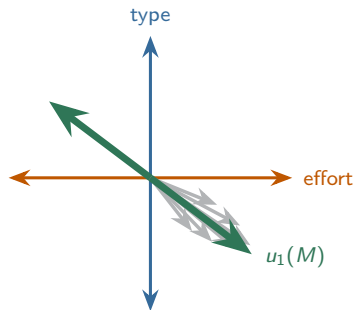
$$p^* \parallel w(\theta).$$

With many types, one signal must serve everyone:

$$\mathbb{E} \left[(p^\top w(\theta))^2 \right] = p^\top M p,$$

$$M = \mathbb{E}[w(\theta)w(\theta)^\top].$$

The relevant aggregation is the second moment, not the mean.



The grey cloud is $\{w(\theta)\}$; its leading direction solves the design problem.

Main result

For small monitoring capacity,

$$V(p) = V(0) + I_g(0) p^\top M p + o(r^2).$$

Therefore

$$\frac{p^*(r)}{r} \longrightarrow \pm u_1(M).$$

where

$$M = \mathbb{E} \left[(I^y)^{-1} \Delta(\theta) \Delta(\theta)^\top (I^y)^{-1} \right].$$

Optimal monitoring targets the leading direction of **information-adjusted incentive needs**.

Statistical versus contracting value

Statistical value

Where is I^y weakest?

Contracting value

Where is M strongest?

In the running example, hold

$$I^y = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

fixed and vary only Δ_θ / Δ_a .

Δ_θ / Δ_a	0.4	1	2
contracting direction	-21°	0°	$+90^\circ$
statistical direction	-32°	-32°	-32°

Statistical informativeness $\neq$ contracting value.

Deconfounding effort and type

Suppose existing monitoring essentially measures

$$a + \theta.$$

For example,

$$y_1 = a + \theta + \varepsilon_1,$$

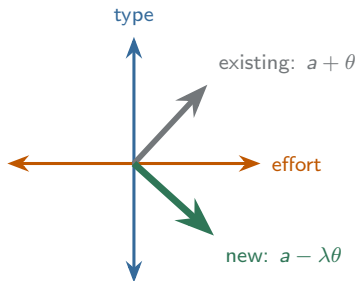
$$y_2 = a + (1 + \delta)\theta + \varepsilon_2, \quad \delta \approx 0.$$

Then the system separates $a + \theta$ well, but barely separates $a - \theta$.

Applying $w = (I^y)^{-1}\Delta$ gives, as $\delta \rightarrow 0$,

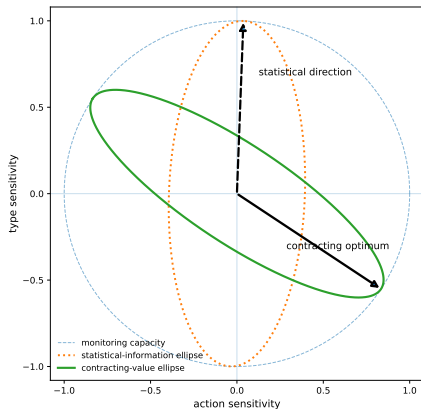
$$\frac{w}{\|w\|} \rightarrow \pm \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (\Delta_a \neq \Delta_\theta).$$

The new signal can use **type** sensitivity to **deconfound effort**.



The new signal points against the confounding in the existing data.

The insurance example, calibrated



Statistical direction

almost pure **type**

$$\phi^{\text{stat}} = 87.9^\circ$$

Contracting optimum

positive **effort**, negative **type**

$$\phi^* = -33.5^\circ$$

The statistical direction delivers only
33% of the available contracting value.

The calibration uses the paper's general multinomial information structure — the Gaussian example was for transparency, the result does not rely on it.

Application: assessment design after AI

$$y = Bx + \varepsilon, \quad x = \begin{pmatrix} a \\ \theta \end{pmatrix}, \quad \varepsilon \sim \mathcal{N}(0, I_2), \quad I^y = B^\top B.$$

a : study effort θ : underlying ability

Homework and exams are already in place; after an AI shock, the professor can redesign the final exam.

Before AI

$$B_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

$$I_0^y = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

After AI

$$B_{\text{AI}} = \begin{pmatrix} 0.7 & 1 \\ 0 & 1 \end{pmatrix},$$

$$I_{\text{AI}}^y = \begin{pmatrix} 0.49 & 0.7 \\ 0.7 & 2 \end{pmatrix}.$$

AI makes homework less informative about effort.

How should the final exam respond?

Same AI shock, different courses

For two equally likely types,

$$w_i = (I^y)^{-1} \Delta(\theta_i), \quad M = \frac{1}{2} \sum_{i=1}^2 w_i w_i^\top, \quad p^* \parallel u_1(M).$$

Suppose the two courses face the same information environment, but have different incentive requirements.

Introductory Microeconomics

Relatively stronger incentives for regular preparation and practice.

Mathematical Economics

Relatively stronger demand for distinguishing analytical ability.

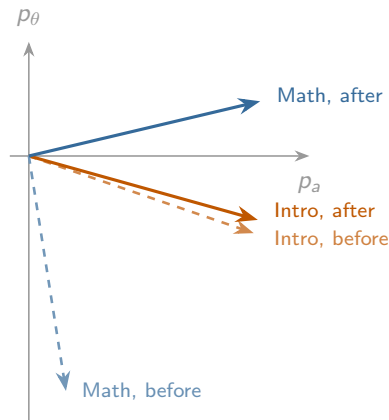
$$\Delta_I(\theta_1) = \begin{pmatrix} 1 \\ 0.4 \end{pmatrix}, \quad \Delta_I(\theta_2) = \begin{pmatrix} 0.8 \\ 0.5 \end{pmatrix}, \quad \Delta_M(\theta_1) = \begin{pmatrix} 0.4 \\ 1 \end{pmatrix}, \quad \Delta_M(\theta_2) = \begin{pmatrix} 0.5 \\ 0.8 \end{pmatrix}.$$

Same I^y , different Δ .

Same information shock, different assessment responses

Normalize $\|p^*\| = 1$ with $p_a^* > 0$. Angles are measured from the effort axis.

Scenario	(p_a^*, p_θ^*)	ϕ^*
Intro, before AI	$(0.946, -0.324)$	-18.9°
Intro, after AI	$(0.963, -0.269)$	-15.6°
Math Econ, before AI	$(0.156, -0.988)$	-81.0°
Math Econ, after AI	$(0.973, 0.230)$	$+13.3^\circ$



Intro: the final mainly supplements effort information, both before and after AI.

Math Econ: before AI, the final is useful for separating effort from ability. After AI, the main missing information is effort itself.

The direction tells us what the *new* assessment should add,

What should the new assessment measure?

For course c ,

$$w_c(\theta) = (I^y)^{-1} \Delta_c(\theta).$$

What is already known?

Existing assessments determine

$$I^y.$$

What information is useful?

The incentive problem determines

$$\Delta_c(\theta).$$

AI changes this information environment by weakening the effort content of homework.

Different courses therefore value the same missing information differently.

What should be measured next depends jointly on what is already measured and what information is needed for incentives.

Re-optimizing the implementation does not change this leading-order characterization.

From local pricing to full incentive compatibility

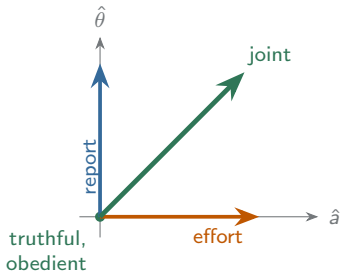
Everything so far used the two local restrictions:

$$\mathbb{E}[v \ell] = \Delta(\theta).$$

Full incentive compatibility is stronger:

$$U(\theta, \theta, a(\theta)) \geq U(\theta, \hat{\theta}, \hat{a}) \quad \forall (\hat{\theta}, \hat{a}).$$

- local pricing controls infinitesimal effort and report deviations;
- full IC controls every joint report–effort deviation;
- the question is whether the leading monitoring direction changes.



Is the eigenvector result an artifact of the local relaxation?

Why the local direction survives

The new signal has sensitivity of order r , so the information it adds is second order:

$$I^s(p) = I_g(0) p p^\top + o(r^2).$$

Hence the monitoring gain is

$$V^{\text{FOC}}(p) - V^{\text{FOC}}(0) = I_g(0) p^\top M p + o(r^2) = O(r^2).$$

Under strict implementation margins, restoring the global joint-deviation constraints costs less than that:

$$V^{\text{FOC}}(p) - V^F(p) = o(r^2) \quad \text{uniformly for } \|p\| \leq r.$$

The $O(r^2)$ monitoring gain dominates the $o(r^2)$ full-IC correction.

The same direction solves the full problem

Therefore the full-IC value has the same leading expansion:

$$V^F(p) = V^F(0) + I_g(0) p^\top M p + o(r^2).$$

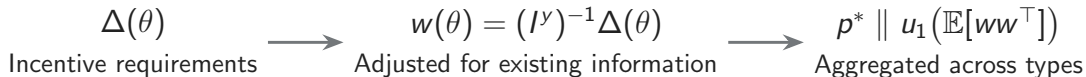
Optimizing over $\|p\| \leq r$ gives

$$\frac{p_F^*(r)}{r} \longrightarrow \pm u_1(M).$$

Local incentive pricing identifies the leading design of the *globally* incentive-compatible mechanism.

If an implementation margin binds at the benchmark, the correction can be order r^2 and the direction can move.

What should the principal measure?



The best thing to measure is not necessarily what is hardest to predict.

It is what is most valuable to *distinguish* for incentives, given what is already measured.

Backup: the timing



y is exogenous

The principal cannot change how informative y is, and she does *not* observe a realization before designing s . At the design stage she has only the statistical structure of y .

The deviation payoff

$$\Pi_p(\theta, \hat{\theta}, \hat{a}) = \mathbb{E}_{\theta, \hat{a}}[\sqrt{t_{\hat{\theta}}(y, s)}] - c(\theta, \hat{a}).$$

Full IC: $(\theta, a(\theta))$ maximizes this over $(\hat{\theta}, \hat{a})$.

Backup: from the Gaussian example to a general q

	Gaussian example	General model
existing measure	$y = Bx + \varepsilon, \varepsilon \sim N(0, \Sigma)$	$y \sim q(y \mid \theta, a), \mathcal{Y} \text{ finite}$
score	$\ell^y = B^\top \Sigma^{-1}(y - Bx)$	$\ell^y = \nabla_{(a, \theta)} \log q$
information	$I^y = B^\top \Sigma^{-1} B$ (constant)	$I^y(\theta, a) = \mathbb{E}[\ell^y(\ell^y)^\top]$
new signal	$s = p^\top x + \eta, \eta \sim N(0, \sigma_s^2)$	$\Pr(s = j) = g_j(p^\top x)$
added information	$I^s = p p^\top / \sigma_s^2$ (exact)	$I^s = I_g(z) p p^\top$ (exact)

What the talk used

Only the *value* of I^y at the benchmark enters Δ , w and M . The Gaussian case makes that value $B^\top \Sigma^{-1} B$ and makes it constant in (θ, a) .

What generality costs

Nothing in the characterization. $I^y(\theta)$ may vary with the type, which is exactly why M averages $w(\theta)w(\theta)^\top$ rather than a single vector.

Backup: the scores, explicitly

The pair (y, s) has likelihood $q(y \mid \theta, a) g_s(z(\theta, a; p))$, so

$$\ell(y, s) = \nabla_{(a, \theta)} \log[q g_s] = \underbrace{\ell^y(y)}_{\text{existing}} + \underbrace{\ell^s(s)}_{\text{designed}}.$$

$$\ell^y = \begin{pmatrix} \partial_a \log q \\ \partial_\theta \log q \end{pmatrix}, \quad \ell^s = \frac{g'_s(z)}{g_s(z)} p.$$

The designed score is a *scalar* times p . That single fact delivers the rank-one structure of I^s .

In the Gaussian example

$$\ell^y = B^\top \Sigma^{-1}(y - Bx) = B^\top \Sigma^{-1} \varepsilon,$$

so $\mathbb{E}[\ell^y] = 0$ and

$$\mathbb{E}[\ell^y (\ell^y)^\top] = B^\top \Sigma^{-1} B,$$

independently of (θ, a) .

Conditional independence of y and s given (θ, a) is what makes the two informations *add*.

Backup: the cheapest contract

Theorem 1

Fix an allocation $\sigma = (a, U)$, a type θ and a design p , and suppose $I = I(\theta, a(\theta); p)$ is nonsingular. Ignoring $v \geq 0$, the unique solution to

$$\min_v \mathbb{E}[v^2] \quad \text{s.t.} \quad \mathbb{E}[v] = \bar{v}_\sigma(\theta), \quad \mathbb{E}[v \ell] = \Delta_\sigma(\theta)$$

is $v^* = \bar{v}_\sigma + \Delta_\sigma^\top I^{-1} \ell$, with value $\bar{v}_\sigma^2 + \Delta_\sigma^\top I^{-1} \Delta_\sigma$.

Proof. Since $\mathbb{E}[\ell] = 0$, write $v = \bar{v} + \lambda^\top \ell + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\mathbb{E}[\varepsilon \ell] = 0$. The requirement gives $I\lambda = \Delta$, so $\lambda = I^{-1}\Delta$, and orthogonality gives $\mathbb{E}[v^2] = \bar{v}^2 + \Delta^\top I^{-1} \Delta + \mathbb{E}[\varepsilon^2]$, uniquely minimized at $\varepsilon = 0$. $\square$

Anything orthogonal to the score span is pure risk: it costs money and satisfies no constraint.

$\sqrt{t}$ is doing real work: it makes the outlay quadratic in promised utility, so incentive cost is a quadratic form in Fisher information.

Backup: why the new signal adds rank-one information

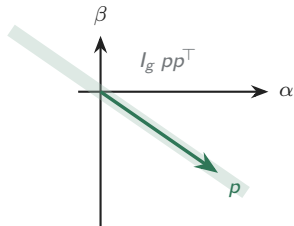
The signal distribution depends on (θ, a) only through the index $z = p^T x(\theta, a)$, so its score is

$$\nabla_{(a,\theta)} \log g_s(z) = \frac{g'_s(z)}{g_s(z)} p.$$

Every realization points along **the same vector** p . With $I_g(z) = \sum_j g'_j(z)^2 / g_j(z)$ and $y \perp s$ given (θ, a) ,

$$I^s = I_g(z) p p^T, \quad \text{tr } I^s = I_g(z) \|p\|^2.$$

Rank one does *not* come from binary support — it comes from the common index.



All the added information lies on one line.

The design chooses *which* line.

At finite r , $I_g(z)$ depends on the state and the direction, so $\|p\| \leq r$ is only *locally* a trace constraint.

Backup: the exact gain from a rank-one update

Lemma 1

Let $A \succ 0$, $\hat{u}$ a unit vector, $\kappa > 0$, $\Delta \in \mathbb{R}^2$, and $w = A^{-1}\Delta$. Then

$$\Delta^T A^{-1} \Delta - \Delta^T (A + \kappa \hat{u} \hat{u}^T)^{-1} \Delta = \frac{\kappa (\hat{u}^T w)^2}{1 + \kappa \hat{u}^T A^{-1} \hat{u}} \geq 0.$$

Apply it at the benchmark with $A = I^y(\theta, a(\theta))$ and $\kappa = I_g(0)r^2$. The denominator is $1 + O(r^2)$, so the reduction in incentive cost for type θ is

$$I_g(0)(p^T w(\theta))^2 + O(\|p\|^3).$$

Two readings: monitoring never hurts, and its gain is bounded above by the “free” value $\kappa(\hat{u}^T w)^2$ — the gap reflects that the new information also raises $\hat{u}^T A^{-1} \hat{u}$.

Backup: from a fixed allocation to the theorem

1. **Fix the allocation.** Proposition 3 and Lemma 1 give, uniformly over $\|p\| \leq r$,

$$V_{\sigma}^{\text{FOC}}(p) = V_{\sigma}^{\text{FOC}}(0) + l_g(0) p^T M_{\sigma} p + O(r^3).$$

2. **Let the allocation move.** Lemma 2 bounds the gain from monitoring by $Cr^2\{\bar{b} - V_{\sigma}^{\text{FOC}}(0)\}$: a substantially suboptimal allocation cannot be rescued by a little measurement. So allocations selected under small designs are benchmark near-optimizers, and Assumption 3 gives $M_{\sigma} \rightarrow M$. An envelope step, not an assumption that the allocation is fixed.
3. **Optimize over directions.** Lemma 3: a uniform quadratic expansion forces $\|p^*\|/r \rightarrow 1$ and selects the top eigenspace of M .

Assumption 3 is an *identification* condition: benchmark near-optimizers must agree asymptotically on M . It does *not* require the benchmark allocation to be unique.

Backup: the confounding threshold in closed form

Suppose I^y is constant across types and the incentive requirements share a direction, $\Delta(\theta) = h(\theta)\delta$ with $\delta = (\delta_a, \delta_\theta)$, $\delta_a, \delta_\theta > 0$. Then M is rank one and

$$\hat{u}_1 \propto (I^y)^{-1}\delta \propto \begin{pmatrix} \iota_\theta \delta_a - \rho \sqrt{\iota_a \iota_\theta} \delta_\theta \\ \iota_a \delta_\theta - \rho \sqrt{\iota_a \iota_\theta} \delta_a \end{pmatrix}.$$

Proposition 4

With $\rho_\theta = \sqrt{\iota_a/\iota_\theta} (\delta_\theta/\delta_a)$ and $\rho_a = \sqrt{\iota_\theta/\iota_a} (\delta_a/\delta_\theta)$: if $\rho_\theta < \rho < \rho_a$ then $\hat{u}_{1,a} > 0$ and $\hat{u}_{1,\theta} < 0$. Moreover $\rho_a \rho_\theta = 1$, so whenever $\rho_\theta < 1$ the range $\rho \in (\rho_\theta, 1)$ is nonempty.

The threshold shows how the two sides interact:

- a larger relative type requirement δ_θ/δ_a raises the confounding needed;
- greater existing precision about effort, ι_a/ι_θ , does the same;
- stronger positive ρ works the other way.

With $\rho = 0$ one gets $\hat{u}_1 \propto (\delta_a/\iota_a, \delta_\theta/\iota_\theta)$ — both positive. Correlation is what reverses the sign.

Backup: the rent channel inside the type requirement

$$\Delta_{\theta}(\theta) = \underbrace{c_{\theta}(\theta, a(\theta))}_{\text{direct effect of type on cost}} + \underbrace{U'(\theta)}_{\text{rent slope}} .$$

$\Delta_a = c_a$ is pinned by the cost technology once the recommended effort is fixed. Δ_{θ} additionally contains the slope of equilibrium utility, so screening creates an extra channel through which the *allocation* determines the demand for information.

The two terms may reinforce or offset. At $U' = -c_{\theta}$ they cancel exactly and $\Delta_{\theta} = 0$.

$\Delta_{\theta} = 0$ removes the *direct* type requirement — not the value of type-sensitive measurement.

Then $w \propto (\iota_{\theta\theta}, -\iota_{a\theta})$, nonzero in the type coordinate whenever the scores are correlated. See the $\Delta_{\theta} = 0$ backup slide.

Backup: the calibration

Primitive	Parameter	Value
Type support	$(\underline{\theta}, \bar{\theta})$	(1, 2)
Effort support	$(\underline{a}, \bar{a})$	(0.05, 2.75)
Cost heterogeneity	γ	0.20
Marginal benefit	b_a	4.25
Benefit–type interaction	$b_{a\theta}$	0.40
Performance index 1	$(\lambda_{1a}, \lambda_{1\theta})$	(1.00, 1.02)
Performance index 2	$(\lambda_{2a}, \lambda_{2\theta})$	(−0.55, 1.18)
Monitoring center	(a_0, θ_0)	(1.20, 1.50)

$$\theta \sim U[1, 2]; \quad c(\theta, a) = \frac{a^2}{2[1 + \gamma(\theta - \underline{\theta})]};$$

$$b(\theta, a) = [b_a + b_{a\theta}(\theta - \theta_0)] a.$$

Existing record. $y \in \{0, 1, 2\}$, multinomial logit

$$q_j(\theta, a) = \frac{\exp\{\eta_j\}}{1 + \exp\{\eta_1\} + \exp\{\eta_2\}},$$

$$\eta_j = \lambda_{ja}a + \lambda_{j\theta}\theta.$$

The two outcomes respond *differently* to effort and type, so the record is informative about both but measures neither separately.

New monitor. Binary logit in the index

$p_a(a - a_0) + p_\theta(\theta - \theta_0)$;
at $p = 0$ it is a fair coin, so $I_g(0) = 1/4$.

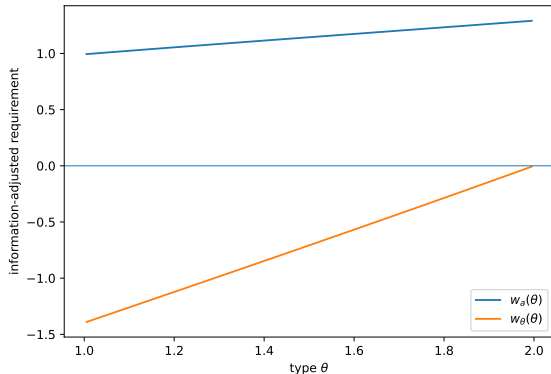
Result. $\nu_1(M) = 1.8255$;

$\hat{u}_1 = (0.834, -0.552)$, $\phi^* = -33.5^\circ$;

$\hat{u}^{\text{stat}} = (0.037, 0.999)$, $\phi^{\text{stat}} = 87.9^\circ$;

value = $I_g(0)\nu_1(M)r^2 = 0.4564 r^2$.

Backup: the information-adjusted requirements by type

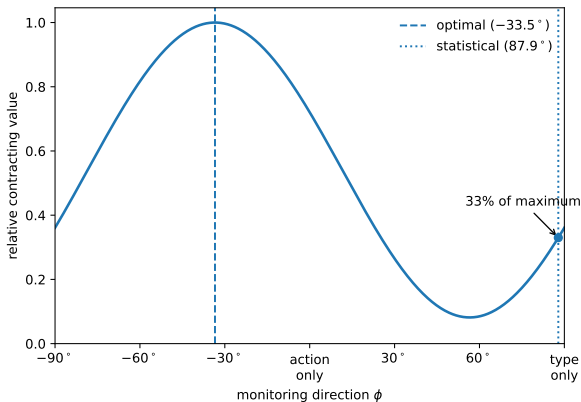


The two components of $w(\theta) = I^y(\theta, a(\theta))^{-1} \Delta(\theta)$ at the uninformative benchmark.

- $w_a(\theta) > 0$ throughout — effort information is always wanted.
- $w_\theta(\theta) < 0$ throughout, approaching zero for high types.

The sign pattern is uniform across types, which is why the aggregate direction inherits it.

Backup: contracting value by direction



For $u(\phi) = (\cos \phi, \sin \phi)^\top$, the curve is $u(\phi)^\top M u(\phi) / \nu_1(M)$.

- Maximum at $\phi^* = -33.5^\circ$.
- The statistical direction, 87.9° , delivers only **33%** of the maximum.
- Pure effort measurement (0°) does much better than the statistical answer.

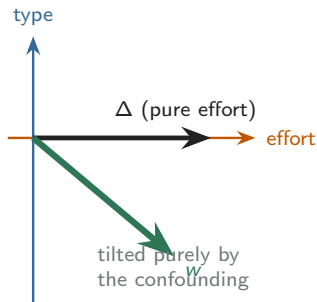
The distinction affects not only *which* direction is chosen but *how much value* scarce monitoring creates.

Backup: a dimension with no direct incentive need

Take the sharpest case: *no* direct type requirement at all, $\Delta_\theta = 0$. The new signal should still respond to type:

$$w = \frac{\Delta_a}{\det I^y} \begin{pmatrix} \iota_\theta \\ -\iota_{a\theta} \end{pmatrix} \neq \begin{pmatrix} * \\ 0 \end{pmatrix} \text{ if } \iota_{a\theta} \neq 0.$$

A type-sensitive measurement can help provide **effort** incentives even when type information has no direct incentive value: the tilt comes entirely from the confounding.



The question is never what a signal measures in isolation, but what the whole system lets the contract *distinguish*.

Backup: when is the statistical criterion right?

Severe confounding

If I^y is close to singular, then

$$(I^y)^{-1} \approx \frac{1}{\lambda_{\min}} v_{\min} v_{\min}^T,$$

so $w \approx v_{\min}$ for *almost any* Δ . The two criteria nearly coincide — the existing system's weakness swamps the incentive weights.

Moderate confounding

If I^y is well conditioned, w depends on Δ and the two criteria come apart. This is the empirically relevant case: several existing measures, none of them redundant.

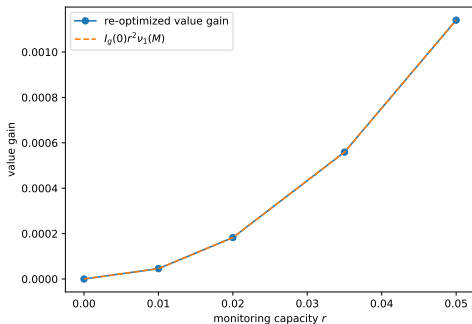
	$\text{cond}(I^y)$	gap between the two directions
near-parallel rows, $\delta = 0.2$	491	0.2°
running example, $\Delta_\theta/\Delta_a = 0.4$	6.9	11°
running example, $\Delta_\theta/\Delta_a = 2$	6.9	122°
calibrated multinomial model	—	121°

The statistical answer is a good approximation *precisely* when the existing system is nearly uninformative in one direction.

Backup: the local approximation is quantitatively accurate

The theorem is *local*: how fast does it degrade?

Along the optimal direction,
 $V(r\hat{u}_1) - V(0) = 0.4564 r^2 + o(r^2)$.



At $r = 0.05$

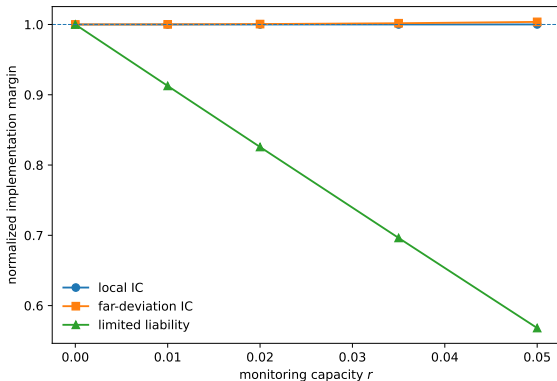
full reoptimization	0.0011404
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quadratic approximation	0.0011409
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Analytically clean *and* quantitatively informative over a nontrivial range.

Over the same range the local IC, far-deviation and limited-liability margins all stay slack.

Backup: implementation margins stay slack



Three margins along the optimal direction, each normalized by its value at $r = 0$:

- **local IC**, measured by $-H_{\max}$;
- **far-deviation IC**, the minimum payoff loss over the nonlocal deviations checked;
- **limited liability**, the minimum transfer.

The first two are essentially flat. Limited liability declines but stays strictly positive — so the value gains are attained in a region where the relaxed allocation really is implementable.

Backup: the assumptions, in one place

- A1 Nondegenerate $I^y(\theta, a) \succeq \varepsilon I_2$ uniformly. More than regularity: without it M can be infinite and the question is ill-posed.
 - A2 LL slack The unconstrained cheapest contract stays bounded away from zero near the benchmark. *Substantive* — where $v \geq 0$ binds, the clean quadratic form is lost.
 - A3 Well-posed $V^{\text{FOC}}(0)$ is attained, and all benchmark near-optimizers share the same limit M . An *identification* condition; does not require uniqueness of the benchmark allocation.
 - A4 Strict local $R_{aa,0} \leq -\eta$ and $\partial_{\hat{\theta}}^2 G_0^L \geq \eta$ at the benchmark, stable along the selected mechanisms.
 - A5 Robust global A uniformly positive payoff gap on deviations at distance $\geq \delta$ from the truthful-obedient pair, preserved under small designs.
- A4–A5 are what the full-IC argument needs. If a margin *binds*, restoring feasibility can cost $O(r^2)$ and differently across directions — and then the direction can move.

Backup: extensions and what is not established

Away from the benchmark

With non-negligible monitoring, the allocation, Δ and I^y all move with the technology, so the optimal direction can rotate as capacity grows. $u_1(M)$ answers “what first”, not “what system”.

Multidimensional monitoring

With several signals the design object is a psd information increment. If capacity can be concentrated, the leading eigenvector still wins; under diversification constraints the whole *spectrum* of M matters.

Dependent signals

Conditional independence makes Fisher informations add. With dependence the relevant object is the *incremental* score after accounting for y , and the rank-one update must be replaced.

Binding implementation

If a local or global margin binds at the benchmark, the design must also be priced against the binding constraint. A natural extension attaches shadow values to it.

Also maintained: $\sqrt{t}$ preferences, a common-index technology, and the $r \downarrow 0$ limit.

Backup: where this sits

- Informativeness** Holmström (1979), Shavell (1979), Gjesdal (1982), Kim (1995), Jewitt (1988), Chaigneau–Edmans–Gottlieb (2019) rank *given* information systems. We instead ask which *direction* a marginal unit should load on, and price it through the Fisher information of the joint experiment.
- Monitoring design** Georgiadis–Szentes (2020) design a signal about a hidden effort, organized around a one-dimensional score. Sun–Zhang (2026) bound the coarseness of an optimal experiment. Adding a hidden type makes the score a two-vector, so the cost becomes a Fisher information *matrix* and the optimum is a direction.
- Sensitivity and precision** Banker–Datar (1989), Feltham–Xie (1994), Holmström–Milgrom (1991), Baker (2002). Our $w = (I^y)^{-1}\Delta$ is the multivariate counterpart — the adjustment is by a matrix, which is exactly what generates the sign reversal.
- AS + MH** Laffont–Tirole (1986), Picard (1987), Gottlieb–Moreira (2022), Castro-Pires–Chade–Swinkels (2024), Myerson (1982). We do not solve the joint allocation problem; we price its two local margins through one information matrix.
- Muddled signals** Frankel–Kartik (2019, 2022), Ball (2025), Hörner–Lambert (2021) take loadings as given or choose them for predictive accuracy. We choose them to minimize the cost of providing incentives.